

Contents

✓ **Methanol** (New Engines)

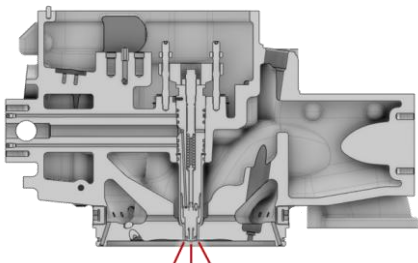
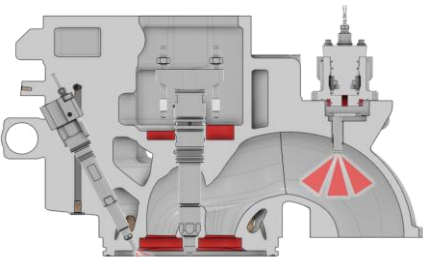
✓ **Ammonia** (New Engines)

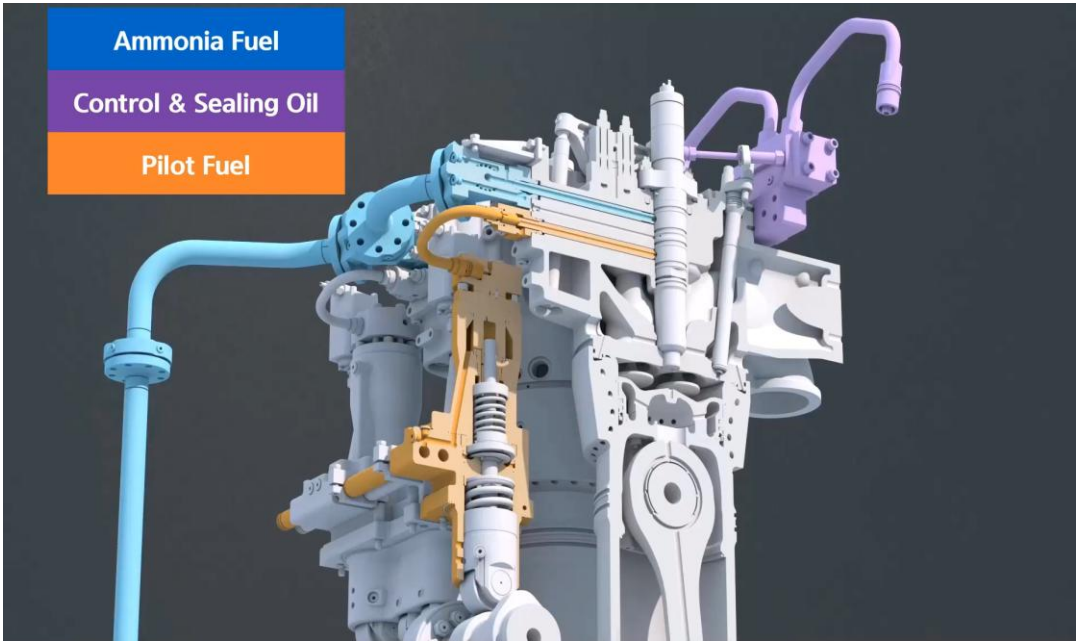
✓ **LNG** (Methane Slip Reduction Improvement)



Diesel Cycle Improves Combustion Reliability

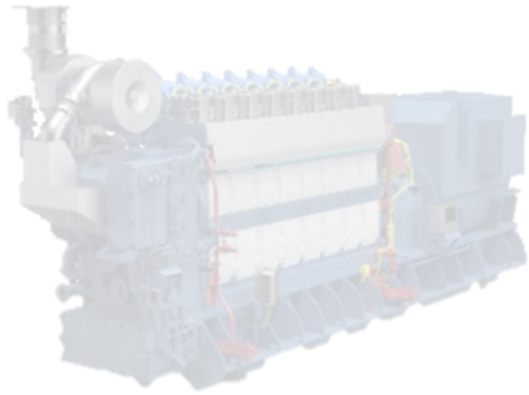
Diesel cycle is adopted for HiMSEN ammonia engine to ensure high efficiency and low emission!

Item		Diesel Cycle	Otto Cycle
Concept		 High Pressure & Liquid Injection	 Low Pressure & Vapor Injection
Output De-rating		None	Un-avoidable
Efficiency		High	Low
Emission	N ₂ O	Low	High
	NH ₃	Low	High (Blow-by, Slip)
	NOx	Relatively high (handled by SCR)	Relatively low



“Nitrous Oxide (N₂O) with global warming potential of 298 over 100-year is one of the most potent green house gases.”

HiMSEN Ammonia Specification



H32DF-LM	
Combustion	Diesel cycle
Bore	320 mm
Stroke	400 mm
Speed	720 rpm
Cyl. config.	6,7,8,9 (inline)
Cyl. output	500 kW
Eng. output	3.0...4.5 MW

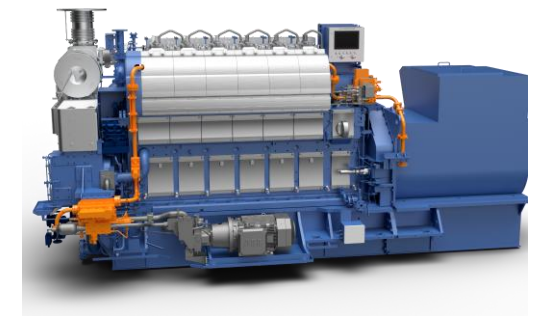
H32CDF-LA	
Combustion	Diesel cycle
Bore	320 mm
Stroke	450 mm
Speed	720 rpm
Cyl. config.	6,7,8,9 (inline)
Cyl. output	600 kW
Eng. output	3.6...5.4 MW

TAT was completed just before.

H22CDF-LM	
Combustion	Diesel cycle
Bore	220 mm
Stroke	330 mm
Speed	900 rpm
Cyl. config.	6,7,8,9 (inline)
Cyl. output	240 kW
Eng. output	1.4...2.2 MW

H22CDF-LA	
Combustion	Diesel cycle
Bore	220 mm
Stroke	330 mm
Speed	900 rpm
Cyl. config.	6,7,8,9 (inline)
Cyl. output	240 kW
Eng. output	1.4...2.2 MW

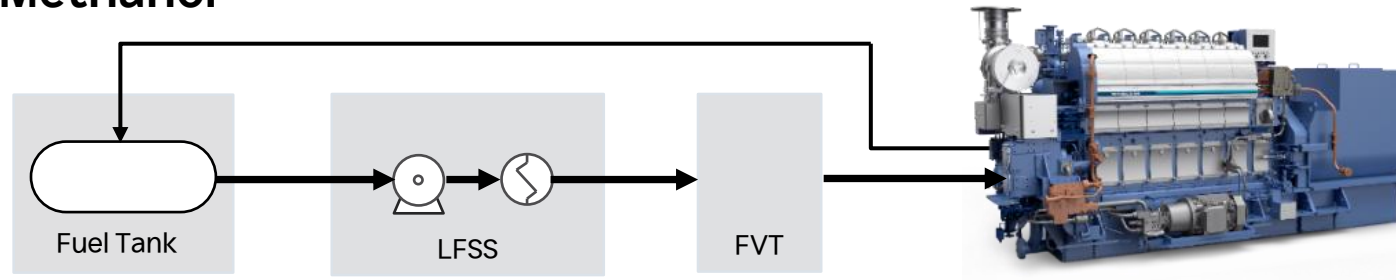
✓ **New in 2024**



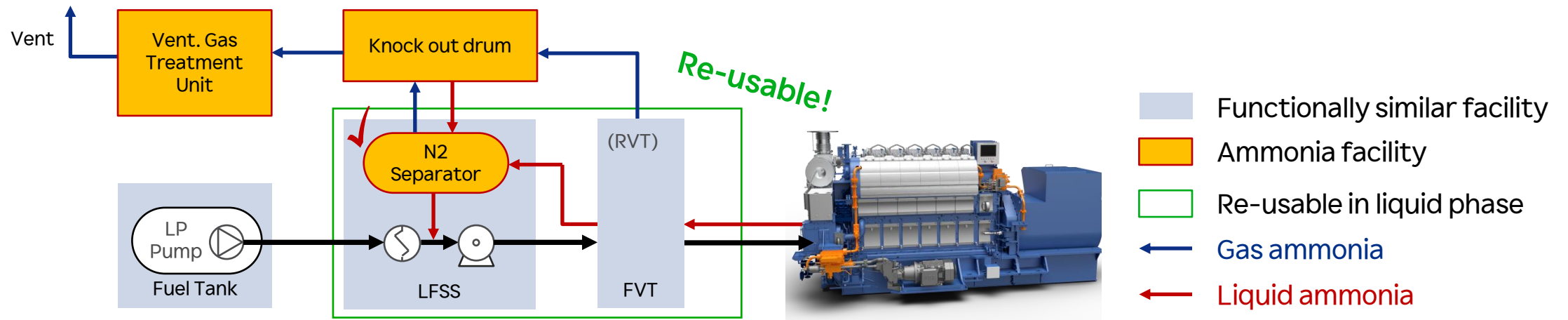
Ammonia Supply System

For using ammonia, a treatment system is necessary

Methanol



Ammonia



H22CDF-LA, Type Approval Test (30th Sep. to 2nd Oct. 2024)

Classification societies : ABS, BV, DNV, KR, LR, NK, RINA



H22CDF-LA Document for Type Approval HD HYUNDAI HEAVY INDUSTRIES

Results of Type Approval Test

(1st day - Diesel mode)
(2nd day - Ammonia mode)
(3rd day - Overhaul & Inspection)

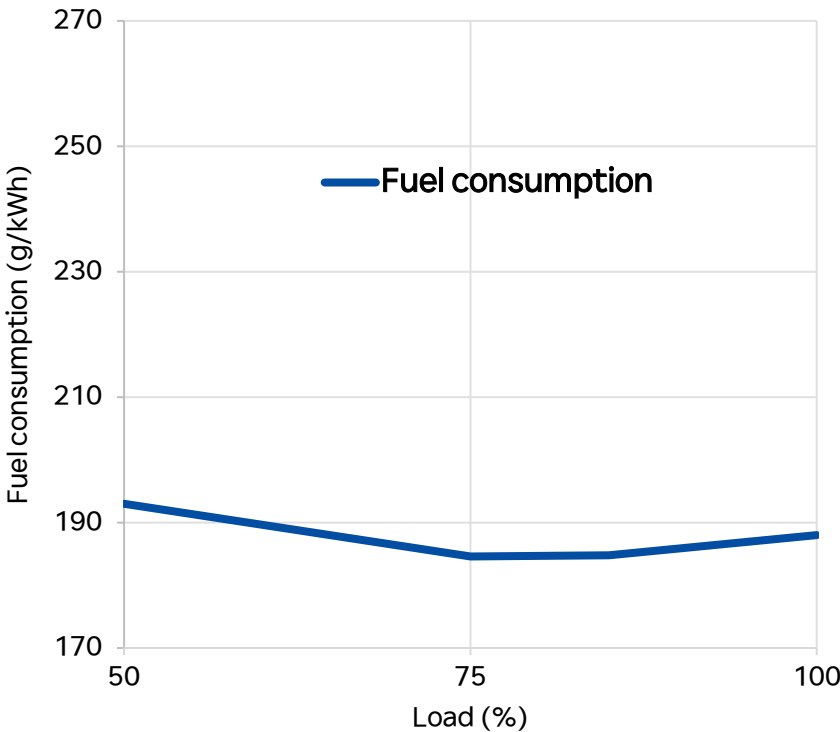
Sep. 30th ~ Oct. 2nd, 2024

Witnessed by the Surveyors

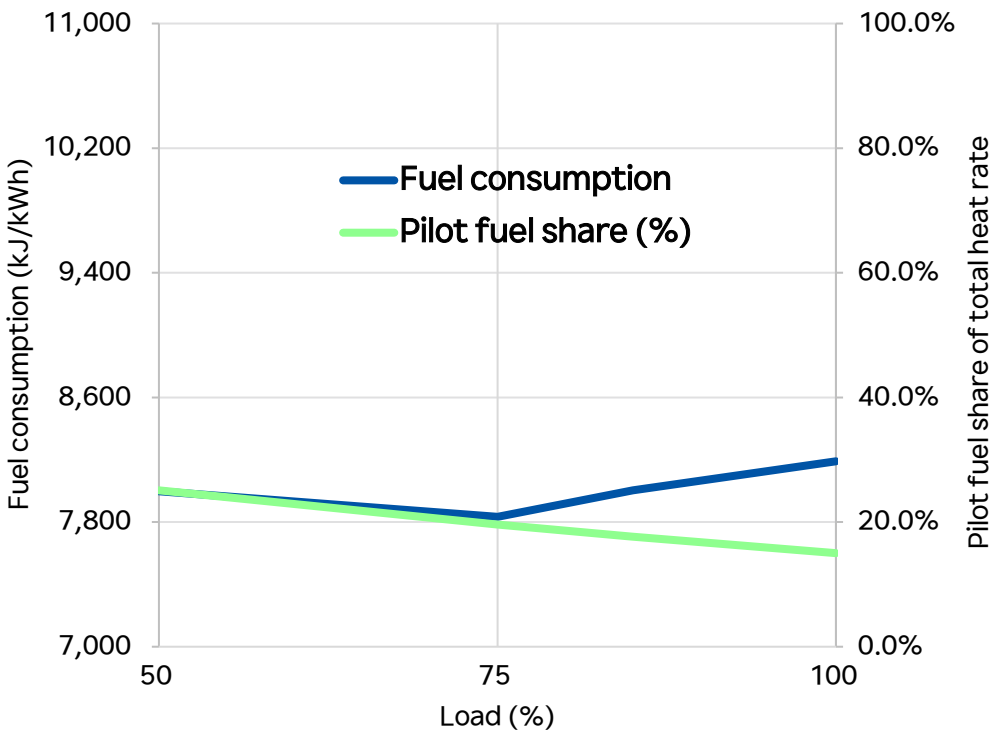
ABS <i>[Signature]</i> 22. October 2024 American Bureau of Shipping <i>[Signature]</i> 2024-10-02 Det Norske Veritas <i>[Signature]</i> 02. Oct. 2024 Lloyd's Register <i>[Signature]</i> 02/10/24 RINA Società per Azioni	<i>[Signature]</i> TAC No. 78019/A0 BV Bureau Veritas <i>[Signature]</i> KR C. 4/ 3/ Oct. 2024 Korean Register <i>[Signature]</i> Nippon Kaiji Kyokai
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H22CDF-LA, Fuel Consumption

Optimization for main operating loads with low pilot fuel share in methanol mode



Diesel mode



Ammonia mode

- NOTE
- Based on ISO conditions
 - Exh. gas back pressure max 300mmWC
 - Engine driven pumps attached (LO, LTCW, HTCW)
 - Fuel oil and pilot fuel based on MDO, LCV 42,700kJ/kWh
 - Ammonia based on LCV 18,600kJ/kg

H32CDF-LA, Type Approval Test (29th Sep. to 2nd Oct. 2025)

Classification societies : ABS, BV, DNV, KR, LR, NK, RINA



H21/32
The first of HIMSSEN engine
2002

H32/40(V)
Advancing toward engine lineup
2006

H35DF(V)
HIMSSEN's first dual fuel engine
2012

H46/60(V)
HIMSSEN's maximum power diesel engine
2013

H32DF-LM
HIMSSEN's first methanol engine
2022

H54DF(V)
World's most powerful dual fuel engine
2019

H32C(V)
HIMSSEN's first two stage turbo charging system engine
2019

H21C
Step forward for high efficiency
2015

HiMSSEN H32CDF-LA(Ammonia)
Type Approval Test
2025. 09. 29 ~ 2025.10. 02

HD HYUNDAI HEAVY INDUSTRIES | ENGINE & MACHINERY

ECO MOBILITY DYNAMICS

THANK YOU